

Circumference, Area & Volume

US Common Core – High School Geometry

Domain: Geometric Measurement and Dimension (G-GMD.A) | Circles (G-C.B)

1 Degrees and Radians

Angle Measure

An angle can be measured in **degrees** ($^{\circ}$) or **radians** (c , or simply as a multiple of π):

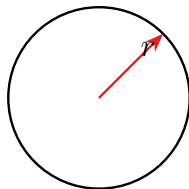
$$\pi = 180^{\circ}, \quad 2\pi = 360^{\circ}, \quad \frac{\pi}{2} = 90^{\circ}.$$

2 Circles

Circumference and Area

For a circle of radius r :

$$\text{Circumference} = 2\pi r, \quad \text{Area} = \pi r^2.$$



Arc Length

For a central angle θ measured in **radians**, the arc length is

$$s = r\theta.$$

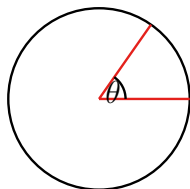
Sector Area

For a central angle θ measured in **radians**, the sector area is

$$A_{\text{sector}} = \frac{1}{2}r^2\theta.$$

For θ measured in **degrees**, equivalently:

$$s = \frac{2\pi r\theta}{360^{\circ}}, \quad A_{\text{sector}} = \frac{\pi r^2\theta}{360^{\circ}}.$$



3 Squares and Rectangles

Formulas

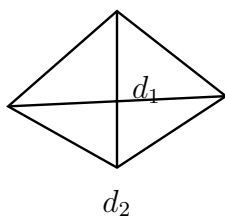
- Area of a square with side a : a^2 .
- Diagonal of a square with side a : $a\sqrt{2}$.
- Volume of a cube with side a : a^3 .
- Area of a rectangle with length ℓ and width w : ℓw .

4 Quadrilaterals with Diagonals

Area of a Rhombus or Kite

For a rhombus or kite with diagonals d_1 and d_2 :

$$\text{Area} = \frac{1}{2} d_1 d_2 = \frac{d_1 d_2}{2}.$$

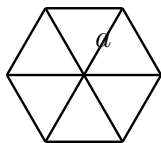


5 Regular Polygons

Area of a Regular Polygon

For a regular polygon with apothem a , n sides, each of length s :

$$\text{Area} = \frac{1}{2} a (ns) = \frac{1}{2} \times \text{apothem} \times \text{perimeter}.$$

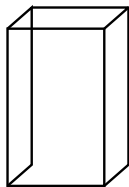


6 Prisms and Cylinders

Volume of a Prism

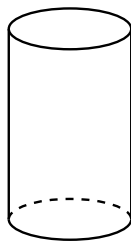
$$V = B \times h$$

where B is the area of the base and h is the height.



Volume of a Cylinder

$$V = B \times h = \pi r^2 h$$



7 Pyramids and Cones

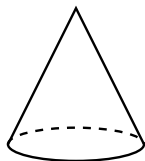
Volume of a Pyramid

$$V = \frac{1}{3} B \times h = \frac{Bh}{3}$$

Cone

$$\text{Surface Area} = \pi r^2 + \pi r \ell \quad \text{Volume} = \frac{1}{3} \pi r^2 h = \frac{\pi r^2 h}{3}$$

where ℓ is the slant height.



8 Spheres

Sphere

$$\text{Volume} = \frac{4}{3}\pi r^3$$

Hemisphere (Half-Sphere)

$$\text{Curved Surface Area} = 2\pi r^2 \quad \text{Volume} = \frac{2}{3}\pi r^3$$

Note: The source notes give the hemisphere's surface area as $\pi r^2/2$. Since a full sphere's surface area is $4\pi r^2$, half of that curved surface is $2\pi r^2$, not $\pi r^2/2$ – the value above has been corrected accordingly. (The flat circular base adds a further πr^2 if the total surface, not just the curved part, is wanted.)

